

4.02 Complex Numbers

Complex numbers and their basic arithmetic are introduced, including in modulus-argument form.

They are used to solve polynomial equations with real coefficients and to define loci on the Argand diagram.

De Moivre's theorem is used to develop trigonometrical relationships. Roots of complex numbers are used to help to solve geometrical problems using an Argand diagram.

4.09 Polar Coordinates Curves defined in polar coordinates are explored, including finding the area enclosed by a curve.

4.10 Differential Equations

The work in A Level Mathematics is extended to include the integrating factor method for first order differential equations. The general 2nd order linear differential equation is solved, including SHM and damped oscillations. Simple systems of linear 1st